Title: Rethinking Over-the-air Reliability: A Energy-harvesting Sensor Nodes Perspective

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Abstract:

Over-the-air reliability remains a critical challenge for wireless sensor networks, particularly in smart home automation. This paper proposes a framework built on energy-harvesting sensor nodes that explicitly accounts for device heterogeneity management. We evaluate the framework on the TON_IoT telemetry dataset and compare it against 5 baseline methods. Experimental results show that the proposed approach improves energy per transmitted bit by 12.5% while keeping packet delivery ratio within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of wireless sensor networks has changed how researchers and practitioners approach over-the-air reliability, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Despite considerable progress, existing solutions for over-the-air reliability tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in detection rate frequently come at the expense of packet delivery ratio, forcing designers to accept trade-offs that are poorly understood. This paper addresses these gaps by studying how energy-harvesting sensor nodes interact with MQTT-based messaging layers when applied to over-the-air reliability.

Our work is motivated by the growing demand for smart home automation, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of over-the-air reliability, a practical design built on energy-harvesting sensor nodes, and an empirical evaluation on the TON_IoT telemetry dataset. Understanding device heterogeneity management is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Chen et al. (2012) introduced an early framework based on energy-harvesting sensor nodes, reporting a 7.1% improvement in end-to-end delay over previous baselines. Subsequent work by Alvarez et al. (2012) extended this line of research to smart home automation, although the evaluation was restricted to small-scale settings. A survey by Banerjee et al. (2009) categorised existing approaches to over-the-air reliability into three families and highlighted the lack of standard benchmarks. Researchers working on wireless sensor networks have explored MQTT-based messaging layers as an alternative, with Okafor et al. (2014) noting that the approach scales poorly beyond 16 components. Banerjee et al. (2008) analysed device heterogeneity management and showed that purely heuristic solutions cannot guarantee consistent energy per transmitted bit.

More recently, Sharma et al. (2009) combined energy-harvesting sensor nodes with MQTT-based messaging layers, achieving a 11.8% reduction in overhead on the TON_IoT telemetry dataset. Several studies have criticised the reliance on synthetic workloads, arguing that results on the TON_IoT telemetry dataset may not generalise to smart home automation. In contrast to these efforts, our work treats over-the-air reliability and device heterogeneity management jointly rather than as independent concerns. Industry reports corroborate

these academic findings, pointing to over-the-air reliability as a persistent obstacle for smart home automation.

3. Methodology

We formalise over-the-air reliability as a constrained optimisation task in which the objective is to maximise energy per transmitted bit while keeping packet delivery ratio within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on energy-harvesting sensor nodes, and a validation layer that incorporates MQTT-based messaging layers. All experiments were conducted on the TON_IoT telemetry dataset, which provides 16 labelled instances or configurations covering a broad range of operating conditions. Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data.

The core routine applies energy-harvesting sensor nodes to reduce the effect of device heterogeneity management, followed by a calibration step that stabilises packet delivery ratio. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 5 baseline methods, including a conventional approach built on MQTT-based messaging layers and two recent variants reported in the literature. The proposed design improves energy per transmitted bit by 12.5% relative to the strongest baseline and reduces packet delivery ratio by 15.2%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the MQTT-based messaging layers component reduces energy per transmitted bit by approximately 8.2%, confirming its contribution. A sensitivity analysis over 128 parameter settings shows that performance remains stable, with variation in energy per transmitted bit below 14.3%. Threats to validity include the limited diversity of the TON_IoT telemetry dataset and the possibility that deployment in smart home automation introduces conditions not captured in our setup.

4. Conclusion

This paper investigated over-the-air reliability through the lens of energy-harvesting sensor nodes, and demonstrated that a carefully designed pipeline can improve energy per transmitted bit by 12.5% on the TON_IoT telemetry dataset. The findings suggest that wireless sensor networks will continue to benefit from closer integration of energy-harvesting sensor nodes and MQTT-based messaging layers, particularly in smart home automation. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of device heterogeneity management. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for over-the-air reliability.